SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Bhavatharani - H (192126056) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 6marks	Q2 6marks	Q3 6marks	Q4 6marks	Q5 6marks
Problem Identification	20%	2	2	2	2	2
Debugging	25%	2	2	2	2	2
Optimization	20%	2	2	2	0	2
Analysis	20%	0	0	0	0	0
Code Quality	15%	0	0	0	0	0

28/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

SIMATS ENGINEERING

ASSESSMENT TOOL 3 – CO2 AT3

Debugging & Optimisation Task

Student Name	Bhavatharani H
Register Number	1921260256
Course Code	CSA0709
Course Name	Computer Networks
Course Outcome	CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using a simulation tool, for efficient network design and topology (BL3)
Tools	Cisco Packet Tracer, Wireshark
Marks	5 questions × 6 marks = 30 marks

Question 1: IPv4 Workstation Configuration (6 marks)

Given: IP 192.168.10.65, Mask 255.255.255.192 (/26), Gateway 192.168.10.1. The workstation cannot reach hosts in 192.168.10.0/26.

1. Subnet of the configured IP

A /26 mask has 6 host bits, so the block size is 256 – 192 = 64. The subnets in 192.168.10.0 are .0, .64, .128 and .192. The address .65 falls in the block starting at .64, so it belongs to **192.168.10.64/26**, not to 192.168.10.0/26.

Item	192.168.10.0/26 (target subnet)	192.168.10.64/26 (workstation's subnet)
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62

2. Validity of the IP address and gateway

- **IP 192.168.10.65:** a valid host address (the first usable address of 192.168.10.64/26), but it is in the *wrong subnet* for the intended LAN.
- **Gateway 192.168.10.1: invalid** for this host. It lies in 192.168.10.0/26, while the host is in 192.168.10.64/26. A default gateway must be in the same subnet as the host's interface, otherwise ARP for the gateway never succeeds.

3. Configuration errors identified

- Subnet mismatch: host IP (.64/26) and gateway/peers (.0/26) are in different subnets.
- Because the subnets differ, the workstation treats 192.168.10.x (x < 64) as remote and sends the traffic to a gateway it cannot reach at Layer 2, so communication fails.

4. Optimised configuration

Option A (preferred – place the host in the intended subnet 192.168.10.0/26):

Parameter	Corrected value
IP address	192.168.10.10 (any unused address in .2 – .62)
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1 (first usable address, router interface)

Option B (keep the host in 192.168.10.64/26): IP 192.168.10.65, mask 255.255.255.192, and the gateway changed to a router interface in the same subnet, e.g. **192.168.10.126**. The IP .65 is then moved to a different address such as .66 if the router uses .65.

5. Total usable host addresses after correction

Usable hosts = $2^n - 2 = 2^6 - 2 = 64 - 2 =$ **62 usable hosts** per /26 subnet.

Question 2: IPv6 Addressing Debug (6 marks)

Given: Device A: 2001:db8::ff00:42:8329/64 Device B: 2001:db8:0:0:1::1/64. Both are on the same network but cannot communicate.

1. Full (expanded) notation

Device	Compressed	Expanded
A	2001:db8::ff00:42:8329	2001:0db8:0000:0000:0000:ff00:0042:8329
B	2001:db8:0:0:1::1	2001:0db8:0000:0000:0001:0000:0000:0001

In A, the `::` replaces 3 zero groups (5 groups are written, 8 needed). In B, `::` replaces 2 zero groups (6 groups written).

2. Network prefix of each device (/64 = first 4 groups)

Device	Network prefix (first 64 bits)	Interface ID (last 64 bits)
A	2001:0db8:0000:0000 → 2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000 → 2001:db8::/64	0001:0000:0000:0001

3. Same subnet?

Yes. Both addresses have the identical /64 prefix **2001:db8:0:0::/64**. B's extra '1' sits in the fifth group, which is part of the *interface ID* (bits 65–80), not the subnet ID. It is a common mistake to read it as a subnet ID, which would wrongly suggest a prefix mismatch.

4. Addressing / prefix mismatch

- There is **no prefix mismatch**: both are /64 on 2001:db8::/64, so on-link reachability does not depend on the prefix.
- The addresses are valid but irregular (non-sequential interface IDs, hard to manage and easy to mistype/misread as in the 5th group). Since the prefixes match, the real fault must be checked at other layers: wrong VLAN/port, interface or link down, IPv6 not enabled on one device, ICMPv6/Neighbor Discovery blocked by a firewall/ACL, or a duplicate address.
- Verify with: `show ipv6 interface brief`, `ping 2001:db8::...`, `show ipv6 neighbors`.

5. Optimised configuration

Device	Optimised address	Prefix
Device A	2001:db8:0:1::10	/64 (2001:db8:0:1::/64)
Device B	2001:db8:0:1::20	/64 (2001:db8:0:1::/64)
Router (gateway)	2001:db8:0:1::1	/64

```
ipv6 unicast-routing
interface g0/0
 ipv6 address 2001:db8:0:1::1/64
 no shutdown
! On hosts: static or SLAAC with prefix 2001:db8:0:1::/64
```

6. Number of available host addresses

A /64 leaves 128 – 64 = 64 host bits, so the subnet holds 2^{64} = **18,446,744,073,709,551,616** (about 1.8×10^{19}) addresses. IPv6 has no broadcast address, so effectively all of them are assignable (only the all-zeros Subnet-Router anycast address is reserved).

Question 3: Subnet Allocation Analysis and VLSM Optimisation (6 marks)

Note on assumptions: the question does not list the host requirements per department. The following realistic requirements are assumed for the analysis: **Engineering 50, Sales 25, HR 12, Accounts 10** hosts (97 hosts in total). The method applies to any other figures.

1. Existing allocation of 192.168.1.0/24 (using /26, /27, /28)

Dept	Subnet	Network	Broadcast	Host range	Usable	Needed	Unused / shortfall
Sales	/26	192.168.1.0	192.168.1.63	.1 – .62	62	25	37 unused
Engineering	/27	192.168.1.64	192.168.1.95	.65 – .94	30	50	**20 short**
HR	/28	192.168.1.96	192.168.1.111	.97 – .110	14	12	2 unused
Accounts	/26	192.168.1.128	192.168.1.191	.129 – .190	62	10	52 unused

Totals: usable = 62 + 30 + 14 + 62 = 168; needed = 97; **unused = 91 (excluding the Engineering shortfall of 20)**. The block 192.168.1.112 – .127 (16 addresses) is left stranded between HR and Accounts.

2. Inefficient allocations and mistakes found

- **Engineering (/27, 30 hosts)** is undersized: needs 50 hosts, so it needs at least a /26.
- **Sales (/26)** and **Accounts (/26)** are oversized: a /26 gives 62 hosts for 25 and 10 hosts, wasting 37 and 52 addresses.
- **HR (/28)** is nearly full (only 2 spare), leaving no room for growth.
- Fixed-size thinking (plain subnetting) instead of VLSM: subnets were not sized by requirement, and the largest requirement was not allocated first.
- A 16-address gap (.112 – .127) is fragmented and unused. Note that a /27 cannot start at .112 because /27 blocks must start at multiples of 32 (.0, .32, .64, .96, .128 ...), so the gap cannot be reused for a /27.

3. Optimised VLSM scheme (allocate largest first)

Dept	Hosts needed	Mask	Network	Host range	Broadcast	Usable	Spare
Engineering	50	/26 (255.255.255.192)	192.168.1.0	.1 – .62	192.168.1.63	62	12
Sales	25	/27 (255.255.255.224)	192.168.1.64	.65 – .94	192.168.1.95	30	5
HR	12	/28 (255.255.255.240)	192.168.1.96	.97 – .110	192.168.1.111	14	2
Accounts	10	/28 (255.255.255.240)	192.168.1.112	.113 – .126	192.168.1.127	14	4
Free pool	–	/25	192.168.1.128	.129 – .254	192.168.1.255	126	future use / WAN /30 links

Result: all four departments are now satisfied (Engineering's 20-host shortfall is fixed). Allocated usable addresses fall from 168 to 120 (needed 97, unused 23), and the whole upper half, **192.168.1.128/25** (128 addresses), is free for growth, new VLANs or /30 router-to-router links. Subnets are contiguous, aligned on their block boundaries and non-overlapping, which also keeps route summarisation possible (192.168.1.0/25 summarises all four).

Question 4: IPv6 Addressing Debug (6 marks)

Given: Device A: 2001:db8::ff00:42:8329/64 Device B: 2001:db8:0:0:1::1/64. Both are on the same network but cannot communicate.

1. Full (expanded) notation

Device	Compressed	Expanded
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Yes. Both addresses have the identical /64 prefix **2001:db8:0:0::/64**. B's extra '1' sits in the fifth group, which is part of the *interface ID* (bits 65–80), not the subnet ID. It is a common mistake to read it as a subnet ID, which would wrongly suggest a prefix mismatch.

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- Verify with: `show ipv6 interface brief`, `ping 2001:db8:...`, `show ipv6 neighbors`.

5. Optimised configuration

Device	Optimised address	Prefix
Device A	2001:db8:0:1::10	/64 (2001:db8:0:1::/64)
Device B	2001:db8:0:1::20	/64 (2001:db8:0:1::/64)
Router (gateway)	2001:db8:0:1::1	/64

```
ipv6 unicast-routing
interface g0/0
 ipv6 address 2001:db8:0:1::1/64
 no shutdown
! On hosts: static or SLAAC with prefix 2001:db8:0:1::/64
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A /64 leaves 128 – 64 = 64 host bits, so the subnet holds 2^{64} = **18,446,744,073,709,551,616** (about 1.8×10^{19}) addresses. IPv6 has no broadcast address, so effectively all of them are assignable (only the all-zeros Subnet-Router anycast address is reserved).

Remark: Question 4 uses the same IPv6 addresses as Question 2, so the analysis and corrected addressing are identical. A Packet Tracer test is to ping between the corrected addresses and capture the ICMPv6 Neighbor Solicitation/Advertisement exchange in Wireshark to confirm on-link reachability.

Question 5: Routing Table Debug and Optimisation (6 marks)

Given routing table:

Destination network	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route	10.0.0.1

Packets destined for **192.168.2.100** are not reaching the intended network.

1. Matching entry (longest-prefix match)

Route	Does 192.168.2.100 match?	Prefix length
192.168.1.0/24 (192.168.1.0 – .255)	No (third octet is 1)	–
192.168.2.0/24 (192.168.2.0 – .255)	**Yes**	/24
Default 0.0.0.0/0	Yes (matches everything)	/0

Two routes match; the router chooses the longest prefix, so **192.168.2.0/24 (/24)** wins over the default route (/0).

2. Next-hop router selected

10.0.0.3.

3. Does the destination belong to the intended subnet?

Yes. 192.168.2.100 is within 192.168.2.0/24 (network 192.168.2.0, broadcast 192.168.2.255, hosts .1 – .254).

4. Routing configuration errors

- The router's route *selection* is correct, so the destination is not mis-matched. The problem is in the entry's **next hop**: 10.0.0.3 is either not the router that actually connects to 192.168.2.0/24 (the next hops for the two networks may be swapped or mistyped), or it is unreachable (down, no ARP entry, wrong interface).
- The next-hop router may also lack a route back to the source network (return path missing), so replies are lost.
- The default route is written as 'Default Route' without an explicit 0.0.0.0/0 form and no exit interface is given, so it depends on recursive lookup of 10.0.0.1.
- Check with `show ip route`, `ping 10.0.0.3`, `traceroute 192.168.2.100`, and confirm the physical/Layer-2 link toward 10.0.0.3.

5. Optimised routing table

Destination	Mask	Next hop / exit	Remark
192.168.1.0/24	255.255.255.0	10.0.0.2 (verified reachable)	Specific static route
192.168.2.0/24	255.255.255.0	Correct router for that LAN (confirmed), via exit interface	Fix/verify next hop; add return route on that router
0.0.0.0/0	0.0.0.0	10.0.0.1	Explicit default route for all unknown destinations

```
ip route 192.168.1.0 255.255.255.0 10.0.0.2
ip route 192.168.2.0 255.255.255.0 <verified next hop>
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Further improvements: use specific /24 routes (not broad ones) so longest-prefix matching works predictably; summarise only when both networks share the same next hop (here they use different next hops, so summarising 192.168.0.0/22 would be wrong); remove duplicate or unused entries; add a floating static or a routing protocol (OSPF) for redundancy.

6. Route used when no destination matches

The **default route (0.0.0.0/0) via 10.0.0.1**, for example for 192.168.3.50 or 8.8.8.8.